

POWER BI Course

Instructor: Eng. Raghad AL-Awadeen

Learning Objectives

Understand the ETL process in Power BI

Connect to common data sources

Explore Power Query Editor

Apply cleaning & transformation techniques

Load cleaned data into the data model

Hands-on practice with messy sales dataset

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The ETL Process in Power BI

•ETL Defined:

- Extract:** Pull data from sources (Excel, CSV, SQL, Web)
- Transform:** Clean, shape, and prepare for analysis
- Load:** Send prepared data to Power BI model for relationships & visuals

•Why ETL is important?

- Raw data is messy, inconsistent, and often unusable
- Clean data ensures accuracy in dashboards and reports

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Load vs Transform

Load

- Brings raw data directly into the Power BI model.
- Data remains **untouched**, exactly as it exists in the source.
- Quick way to get data into your report, but **messy or inconsistent data** can cause issues later.

Transform

- Opens **Power Query Editor** before loading.
- Allows you to **clean, shape, and prepare** your data:
 - Remove nulls or duplicates
 - Split or merge columns
 - Change data types
 - Pivot/Unpivot tables
 - Create conditional column

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Power Query Editor – Interface Components:

- **Ribbon** → tools for cleaning, transforming, adding columns
- **Queries Pane (left):** All tables/queries
- **Data Preview (center):** Sample of your dataset
- **Applied Steps (right):** History of transformations
- **Formula Bar:** Shows M code behind each step

Important Note: Power Query is non-destructive → it never changes your source file

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Cleaning – Remove Nulls

- **Why remove nulls?**

- Affects totals, averages, and joins
- Example: Missing Customer ID → order cannot be linked

- **Options:**

- Remove rows with nulls (Remove Rows → Remove Blank Rows)
- Replace nulls with a default (Transform → Replace Values)

- **Common Mistake:** Deleting all nulls without checking → lose valuable data

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Cleaning – Remove Duplicates

- **Why remove duplicates?**

- Double-counting sales or transactions
- Inflates KPIs

- **How:**

- Select column(s) → Remove Rows → Remove Duplicates

- **Best Practices:**

- Define a **unique identifier** (like Order ID)
- If multiple duplicates → investigate before deleting

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Split & Merge Columns

Split Columns:

- Break one column into many using delimiter or fixed width
- Example: Full Name → First Name, Last Name

Merge Columns:

- Combine multiple columns into one
- Example: City + Country → Location

Common Mistake: Forgetting to trim spaces after split/merge

Pivot & Unpivot

Pivot:

- Converts row values into columns (wide format)

Unpivot:

- Converts columns into rows (long format)
- Needed for time-series & clean modeling
- **Best Practice:** Always unpivot time-based columns (Jan-Dec sales)

Conditional Columns

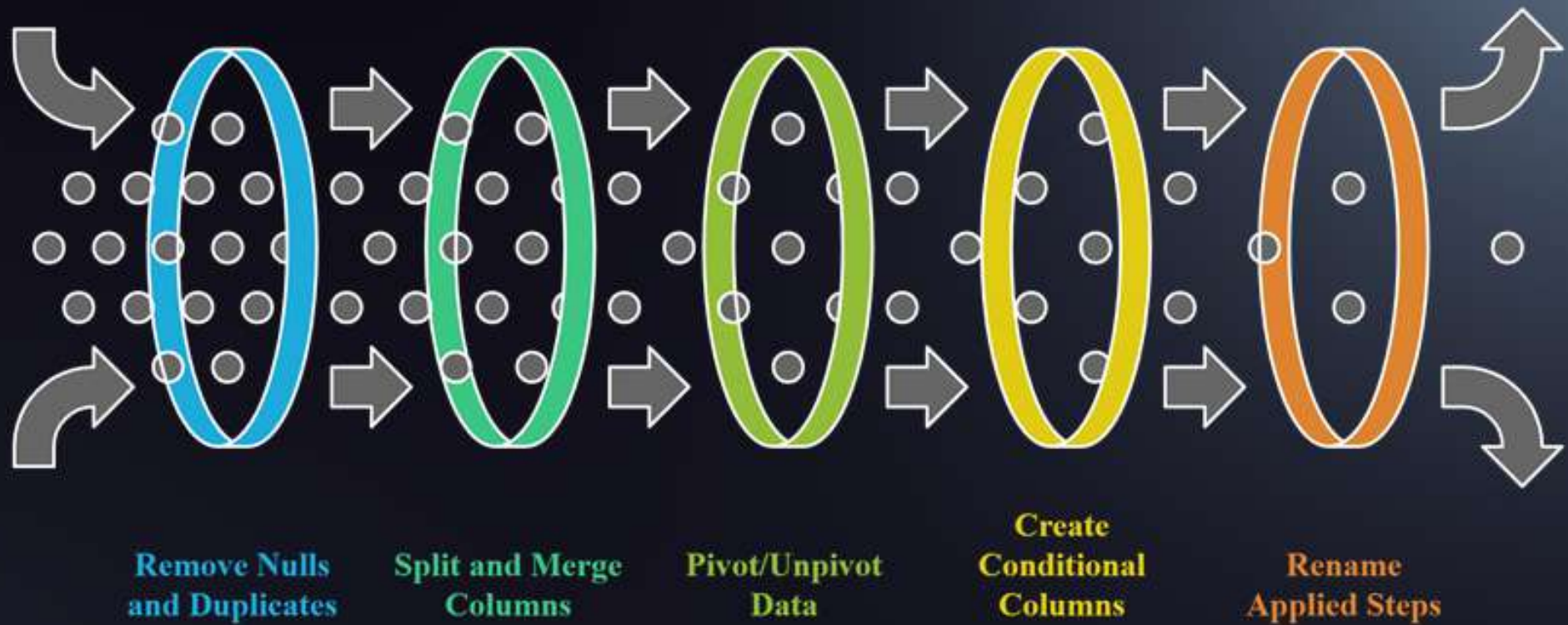
Definition: Add logic-based new columns (like IF in Excel)

Example:

- If Sales > 5000 → "High"
- Else If Sales 2000-4999 → "Medium"
- Else → "Low"

Best Practice: Use ranges for categories, not hard-coding single values

Practice Time – Apply What You’ve Learned



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